chapter: 5

Cauchy's integral formula

i) If z_0 is any arbitrary but fixed point inside the contour c, then

$$f(z_0) = \frac{1}{2\pi i} \oint \frac{f(z)}{z - z_0} dz$$

$$\Rightarrow \oint \frac{f(z)}{z - z_0} dz = 2\pi i \cdot f(z_0)$$

(ii) $$f^n(z_0) = \frac{n!}{2\pi i} \oint_c \frac{f(z)}{(z - z_0)^{n+1}} dz$$

or, $$\oint_c \frac{f(z)}{(z - z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^n(z_0)$$

$f^n \rightarrow$ n-th derivative

Example : 1

Evaluate $\oint_c \frac{e^{3z}}{z - \pi i} dz$ where 'c' is,

a) $|z-1| = 4$

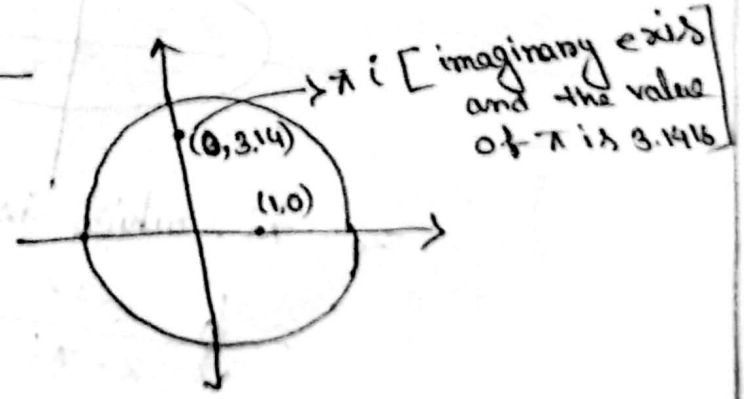
b) $|z-2| + |z+2| = 6$

(a)

$|z-1| = 4$

$\Rightarrow |x + iy - 1| = 4$

$\Rightarrow (x-1)^2 + y^2 = 4^2$



Now,

$z_0 = \pi i$

$\therefore (x-1)^2 + y^2 = 4^2$

where, $(0, \pi)$

$(0-1)^2 + \pi^2 = 4^2$

$10.( ) < 4^2$

$\therefore \oint_c \frac{e^{3z}}{z-\pi i} = 2\pi i\, f(\pi i)$

$\Rightarrow \quad = 2\pi i\, e^{3\pi i} \quad (\because f(z) = e^{3z})$

$= 2\pi i (\cos 3\pi + i \sin 3\pi)$

$= -2\pi i$

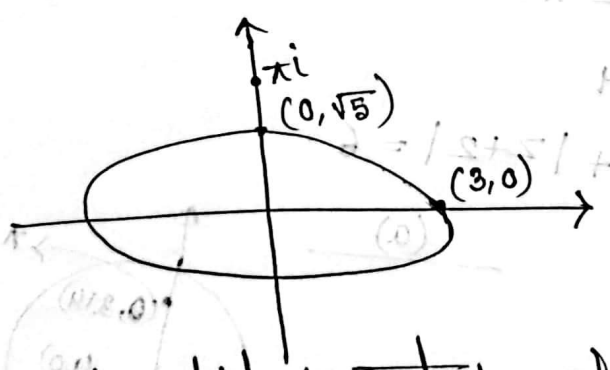
(A)

(b)

$|z-2| + |z+2| = 6$

$\Rightarrow 5x^2 + 9y^2 = 45$

$\Rightarrow \frac{x^2}{3^2} + \frac{y^2}{(\sqrt{5})^2} = 1$



$z_0 = \pi i$; which is outside of c

$\therefore \oint \frac{e^{3z}}{z-\pi i} dz = 0$ (A)

[4] Exp: 2

Evaluate a) $\oint \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} dz$

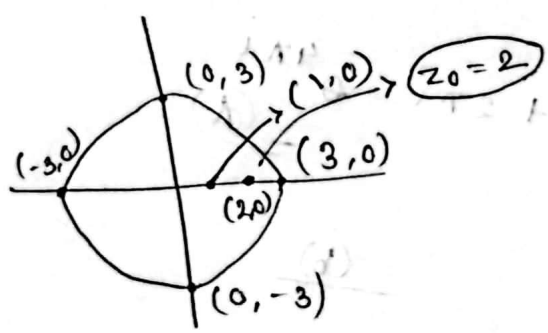
b) $\oint \frac{e^{2z}}{(z+1)^4} dz$

where c is $|z| = 3$

(a)

$|z| = 3$

$\Rightarrow \cancel{x^2 + y^2 = 3^2} \Rightarrow x^2 + y^2 = 3^2$



$\frac{1}{(z-1)(z+2)} = \frac{1}{z-2} - \frac{1}{z-1}$ [∴ partial fraction]

Now,

$\therefore \oint \frac{\sin(\pi z^2) + \cos(\pi z^2)}{z-2} dz - \oint \frac{\sin(\pi z^2) + \cos(\pi z^2)}{z-1} dz$

For 1st integral, $z_0 = 2 = 2 + i.0 \rightarrow (2, 0)$

" 2nd " , $z_0 = 1 + 1 + i.0 \rightarrow (1, 0)$

$$\therefore \oint \frac{\sin(\pi z^2) + \cos(\pi z^2)}{z-2}\, dz = 2\pi i \cdot f(2)$$

$$= 2\pi i \left(\sin(4\pi) + \cos(4\pi)\right)$$

$$= 2\pi i \cdot 1 = 2\pi i$$

$$\therefore \oint \frac{\sin(\pi z^2) + \cos(\pi z^2)}{z-1}\, dz = 2\pi i\, f(1)$$

$$= 2\pi i \left(\sin\pi + \cos\pi\right)$$

$$= -2\pi i$$

$$\therefore \oint \frac{\sin(\pi z^2) + \cos(\pi z^2)}{z-2}\, dz - \oint \frac{\sin(\pi z^2) + \cos(\pi z^2)}{z-1}$$

$$= 2\pi i + 2\pi i = \cancel{0}\ 4\pi i$$

(d)

(b)

Exp: 03

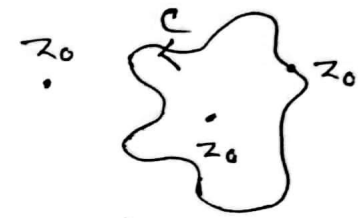
Evaluate $\oint_c \frac{e^z}{(z^2+\pi^2)^2}\,dz$

where,

c is $|z| = 4$

~~cau~~ Cauchy's integral formula:

$$\oint_c \frac{f(z)}{(z-z_0)^{n+1}}\,dz = \frac{2\pi i}{n!} f^n(z_0)$$



Sol^n:

$$\oint \frac{e^z}{(z^2+\pi^2)^2}\,dz = \oint_c \frac{e^z}{\{(z+i\pi)(z-i\pi)\}^2}\,dz$$

$\therefore z^2+\pi^2 = 0$

$\Rightarrow z = \underbrace{\pm i\pi}_{\text{roots}}$

$\underbrace{(z+i\pi)(z-i\pi)}_{\text{factors}}$

$$\therefore \frac{1}{(z+i\pi)^2(z-i\pi)^2} = \frac{1}{(z+i\pi)(z-i\pi)}\left(\frac{1}{(z+i\pi)(z-i\pi)}\right)$$

$$= \frac{1}{(z+i\pi)(z-i\pi)}\left[\frac{1}{(z+i\pi)(-i\pi+i\pi)} + \frac{1}{(z-i\pi)(i\pi+i\pi)}\right]$$

[where $z = -i\pi$]

$z = +i\pi$

$$= \frac{1}{(z+i\pi)(z-i\pi)}\left[\frac{1}{(z+i\pi)(-2i\pi)} + \frac{1}{(2i\pi)(z-i\pi)}\right]$$

$$= \frac{1}{(z+i\pi)(z-i\pi)} \times \frac{1}{2i\pi}\left(\frac{1}{z-i\pi} - \frac{1}{z+i\pi}\right)$$

$$= \frac{1}{2i\pi} \cdot \frac{1}{z+i\pi}\left(\frac{1}{(z-i\pi)^2} - \frac{1}{(z-i\pi)(z+i\pi)}\right)$$

$$= \frac{1}{2i\pi} \cdot \frac{1}{z+i\pi}\left[\frac{1}{(z-i\pi)^2} - \left(\frac{1}{2i\pi}\left\{\frac{1}{z-i\pi} - \frac{1}{z+i\pi}\right\}\right)\right]$$

$$= \frac{1}{2i\pi}\left[\frac{1}{(z+i\pi)(z-i\pi)^2} - \frac{1}{2i\pi} \cdot \left\{\frac{1}{(z-i\pi)(z+i\pi)} + \frac{1}{(z+i\pi)^2}\right\}\right.$$

[cover-up rule]

Ans: $\frac{1}{\pi}$

① $\frac{1}{2\pi i} \oint_C \frac{e^{zt}}{(z^2+1)^2} dz$

Ans: $\frac{1}{2}(\sin t - \cos t)$

$C: |z| = 3$

$\sin t = \frac{e^{it} - e^{-it}}{2i}$

$\cos t = \frac{e^{it} + e^{-it}}{2}$

② $\oint \frac{\cos^2(tz)}{z^3} dz$; $C: |z| = 1$

Ans: $-2\pi i t^2$

$z_0 = 0$

$m = 2$

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